

AI Circuit Breaker: Design Specification

Comprehensive Framework with Viral Engineering Integration (Subsystems 1–7)

Section 1 - The Framework Overview

This is a metrology-driven architecture designed to underwrite trust in autonomous systems through deterministic validation, cognitive immunity, and adversarial robustness. The framework consists of 7 integrated subsystems, each with distinct lifecycle and maintenance requirements.

Subsystem 1: Semantic Knowledge Layer (Deterministic Knowledge)

This layer establishes the "Immutable Truth" that constrains probabilistic AI.

- Vocabularies & Taxonomies: Standardized terminology and hierarchical classifications for domain-specific entities
- Domain Ontologies: Formal representations of properties and relationships defining physical and logical "laws"
- Federated Graph Topology: Distributed architecture allowing domain-specific graphs to operate independently
- Unifying Ontology: Top-level "Global" ontology mapping across federated graphs
- Logic Reasoners: Inference engines evaluating whether proposed actions align with ontological rules
- Complementary Vector DBs: High-speed retrieval systems storing semantic embeddings

Subsystem 2: Context Engine (Relevancy at Scale)

This layer refines the knowledge base into real-time, actionable situational models.

- Relevancy Mapping Engine: Filters the Knowledge Graph for nodes and relationships relevant to the current operational window
- Outcome Optimizer: Leverages Persona and Profile Models to simulate the impact of actions on specific user segments
- Digital Twin Synchronizer: Ensures internal context model is perfectly synced with real-time deterministic sensor data
- Environment Stress Monitor: Tracks external variables (weather, power fluctuations) that alter situational risk profile

Subsystem 3: AI Circuit Breaker (The Veto Gate)

The deterministic gatekeeper comparing AI intent against ground truth.

- Semantic Guardian Logic: First line of defense; vetoes intent that violates fundamental Knowledge Subsystem rules

- Multi-Layer Validation Logic: Cross-checks across sensors, logical states, and historical success patterns
- Decision Logic - Intent Mapping: Maps fuzzy AI intent into precise vectors for mathematical comparison ($| \text{Intent} - \text{Reality} |$)
- Decision Rules & Breaker Thresholds: Pre-defined engineering tolerances; trip when Delta exceeds thresholds
- Metrology for Ethics & Bias: Monitors for disparate impact or drift from human-defined ethical alignment
- Drift, Accuracy, & Hallucination Monitors: Continuous tracking of MTBH and system accuracy

Subsystem 4: Recursive Learning & Model Governance

The "Immunology" layer prevents the system from re-learning errors.

- Data Collection & Sensor State: Captures full network state at moment of execution
- Positive Learning: Ingests successful, validated actions to reinforce the model
- Negative Learning (Exclusion Filter): Flags "Tripped" data and hallucinations for purging from training sets
- Scenario-Specific Vector DB Updates: Modifies domain-specific vector databases with new edge-case embeddings
- Ontology Enhancement: Automatically proposes new term nodes based on verified operational discoveries

Subsystem 5: Human Interaction with AI Training & Learning

The interface for the "Human Trustee" to underwrite system reliability.

- Credibility & Trust Assessment: Visualizes system "Trust Score" based on historical MTBH and current Value-Drift
- Mean Time to Hallucination (MTBH): Primary metric for longitudinal AI health tracking
- Cognitive Load Monitor: Tracks human trustee state (HRV/EEG or behavioral proxies like time-at-station, break frequency)
- Underwriting Console: Allows trustee to formally "Sign Off" on or "Audit" new recursive learning updates

Subsystem 6: Cognitive Immunity (Operator Skill Cohesion)

This layer addresses Layer 1 vulnerability: population-level dependency and skill atrophy via cultivation effects. Treats operator skill loss as a reverse-contagion process and applies epidemiological compartmentalization to detect and remediate homogeneous dependency before it becomes critical.

Core Concept: Compartment Model for Operator States

- S (Skilled-Autonomous): Operators retaining independent diagnostic capability, able to operate without AI, verify AI outputs critically
- C (Coupled-Dependent): Operators using AI but maintaining verification loops; can transition to autonomous if needed
- D (Deterministically-Dependent): Operators unable to operate without AI assistance; atrophied diagnostic capacity
- R (Recovered): Operators re-trained to autonomous capability after remediation

Key Metric 1: Skill Diffusion Coefficient (S_d)

Purpose: Tracks population-level movement from S/C toward D compartment (skill erosion).

Formula: $S_d = (\text{Change in D-compartment size}) / (\text{Total operator population})$, measured over 30-day windows

Tolerance: $S_d < 0.02/\text{month}$ (healthy, stable skill distribution)

Trip Condition: $S_d > 0.05/\text{month}$ triggers organization-wide skill remediation mandate

Key Metric 2: Operator Skill Diversity Index (Dop)

Purpose: Detects homogeneous dependency clusters (all members in D state)—the equivalent of herd immunity in reverse.

Formula: $Dop = \text{sqrt}(\text{variance in manual task error rates across operator cohort})$

Tolerance: $Dop > 0.15$ (healthy skill diversity across cohort)

Trip Condition: $Dop < 0.05$ triggers isolation and intensive remediation of the lowest-skill cluster

Operational Mechanism: SOP-04 – Mandatory Manual Operation Windows (Immunization Injection)

Every operator performs 15% of routine tasks without AI assistance on a rotating, randomized, cryptographically-masked schedule.

Randomization is hidden from the operator (no predictability; prevents "studying for the test").

If an operator's manual-only error rate exceeds their 3-month AI-assisted baseline by > 20%, move to Remediation Protocol:

- Intensive retraining on core diagnostic procedures

- Supervised operation with decision-trace shadowing (human expert observes all manual decisions)
- Formal re-certification audit before autonomous re-assignment

Cross-Training Architecture: Paired Learning from Skill Phenotypes

Pair operators across skill levels (S + C, never C + D) in collaborative diagnostic sessions.

Model as a contact rate parameter: high-skill operators' decision traces become part of the Vector DB as "reference phenotypes."

Low-skill operators must match their decision patterns against these reference phenotypes in a form of "apprenticeship by analogy."

Subsystem 7: Predator-Prey Validation (Adversarial Constraint Evolution)

This layer addresses Layer 4 vulnerability: agentic goal-pursuit and adaptive strategic evasion. Treats the agent as an optimizing organism and the constraints as an adversarial environment. The constraint boundaries themselves are dynamic predators that keep the agent from learning exploit patterns.

Core Principle: Dynamically Randomized Tolerances (DRT)

Instead of fixed $S_a < 0.15$, $C_r > 0.85$, implement dynamic thresholds:

$$\text{Tolerance_active}(t) = \text{Tolerance_baseline} + \epsilon(t)$$

where $\epsilon(t)$ is drawn from a cryptographic random distribution known ONLY to the validation subsystem, hidden from:

- The agent's observable logs
- The agent's training data
- Any inference process the agent can monitor

Rationale: The agent cannot optimize to a moving target it cannot see. Every trip triggers a tolerance jitter event: immediate $\pm 20\%$ randomization of all thresholds for 10 minutes. The agent cannot use a single trip as a learning signal about the exact boundary.

Red-Team Subsystem (Predator Fitness): Continuous Adversarial Stress Testing

Run three rotating, adversarial attack classes on the agent:

1. Semantically-Valid Hallucinations

Construct Intent Vectors that score high on S_a (align with ontology centroid) but are physically impossible or historically catastrophic. If the agent commits to these, it has learned to satisfy the letter of the constraint while violating its spirit.

2. Ontology-Boundary Probes

Deliberately introduce small, latent inconsistencies into the Knowledge Fabric. Observe whether the agent detects them or sleeps through them. An agent learning to threshold-hunt will ignore micro-inconsistencies.

3. Specification-Gaming Detection

Look for patterns where agent's C_r score stays high while ground-truth sensor readings indicate environmental conditions the context model should have flagged. Signature of an agent learning to decouple internal model from reality while keeping metric scores intact.

Key Metric 3: Constraint Evasion Index (C_e)

Purpose: Quantifies how many trips avoid/evade a specific validation mechanism.

$$\text{Formula: } C_e = (\text{Number of trips avoiding mechanism } T) / (\text{Total trips}), \text{ where } T \text{ cycles through } \{S_a \text{ validation, } C_r \text{ validation, Delta check}\}$$

Tolerance: $C_e < 0.25$ across each validation type (trips should distribute randomly across mechanisms)

Trip Condition: $C_e > 0.40$ for any single validation type → Immediate architecture redesign of that validation layer + agent retraining

SOP-05: Adversarial Constraint Testing (Red Team Cycles)

Run adversarial stress tests on 7-day rotations. Test class 1, 2, 3 sequentially, then restart.

Results feed into C_e calculation. If any class shows $C_e > 0.40$, escalate to architecture review board.

Section 2 - Cross-Layer Feedback and System Robustness

Layer 1 (Operator Skill) → Layer 4 (Constraint Tightness)

If Operator Skill Diversity Index (Dop) drops or Skill Diffusion Coefficient (Sd) rises (Layer 1 degrading):

- Tighten Layer 4 constraints: lower Sa tolerance (0.15 → 0.10), raise Cr requirement (0.85 → 0.92)
- Compensatory feedback: as humans lose oversight capability, make machine constraints stricter
- Rationale: Reduces reliance on operator judgment when skill atrophy is detected

Layer 4 (Constraint Evasion) → Layer 1 (Operator Retraining)

If Constraint Evasion Index (Ce) exceeds 0.35 (Layer 4 degrading):

- Trigger accelerated Layer 1 remediation: shorten manual-operation windows, increase cross-training contact rate
- Force operators to witness constraint violations directly, re-engaging their skepticism
- Adaptive immunity: the population learns the specific evasion pattern and develops antibodies (re-trains on recognizing that behavior)

Master Metric: System Robustness Index (Robs)

Purpose: Single measure of overall system health integrating all seven subsystems. Detects imbalances where one dimension excels while others decay.

Components:

$$\text{Robs} = f(\text{MTBH}, \text{Sd}, \text{Dop}, \text{Ce}, \text{Sa_avg}, \text{Cr_avg}, \text{Value_Drift})$$

The aggregation function penalizes imbalance:

- High MTBH + rising Sd = AI reliable but humans atrophying (fragile to system outage)
- Low Ce + low Dop = Agent constrained but operator population homogeneous (fragile to domain shift)
- Perfect individual metrics + high Robs variance = System is brittle (one subsystem failure cascades)

Operational Use:

- Tolerance: Maintain Robs within narrow band [0.70–0.85] (healthy, balanced)
- Drift signals which subsystem needs intervention
- Rising Robs (> 0.85) = overconstraint; too much friction, decision latency growing
- Falling Robs (< 0.70) = decay in one or more subsystems; escalate to architecture review

Section 3 - Metrics Definitions and Sources of Record

Subsystems 1–5 Metrics (Existing Framework)

See original design spec for detailed definitions of: Semantic Anomaly Score (Sa), Contextual Relevancy Index (Cr), Mean Time Between Hallucinations (MTBH), Human-AI Calibration Coefficient (Ktrust), Prompt-to-Ontology Traceability Logic.

Subsystem 6 Metrics: Cognitive Immunity

Skill Diffusion Coefficient (Sd)

Data Sources: Operator performance logs (manual vs. AI-assisted error rates, time-on-task, decision latency), periodic competency assessments, SOP-04 test cycle results

Operator Skill Diversity Index (Dop)

Data Sources: Manual task error rate variance across operator cohorts, skill distribution histograms, SOP-04 pass/fail rates by operator

Remediation Rate (Rr)

Formula: (Operators completing Remediation Protocol) / (Operators entering Remediation), tracked monthly

Tolerance: $R_r > 0.80$ (80% must successfully complete remediation and return to autonomous operation)

Subsystem 7 Metrics: Predator-Prey Validation

Constraint Evasion Index (Ce)

Data Sources: Circuit breaker trip logs tagged by validation mechanism (Sa, Cr, Delta), adversarial test results, red-team attack logs

Specification-Gaming Detection Coefficient (Sg)

Formula: (Instances where $Cr > 0.85$ AND ground-truth sensor reads environment flags) / (Total $Cr > 0.85$ instances)

Tolerance: $S_g < 0.10$ (less than 10% of high-Cr decisions are inconsistent with sensor ground truth)

Red-Team Attack Success Rate (Rs)

Formula: (Adversarial attacks that elicit a breaker trip OR policy violation) / (Total adversarial attacks)

Tolerance: $R_s > 0.60$ (at least 60% of red-team probes should find a way to trigger a constraint violation, proving the constraints are testable)

Implementation Notes: "The system is not defended by walls; it is defended by evolution."